

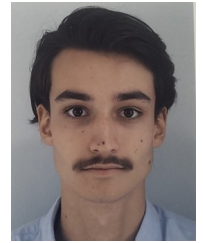
Victorien THOMAS

Aerospace engineering student — Simulation & modelling of space systems

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Seeking a PhD in space engineering — available from July 2027



EDUCATION

Master in Space and Aeronautical Engineering (MASE) — double degree <i>UPC – ESEIAAT · Terrassa, Spain</i>	2026 – 2027
Exchange semester <i>Universidad EIA · Medellín, Colombia</i>	Jul. – Nov. 2025
Master's-level Engineering Degree in Aerospace — Space Systems major <i>IPSA · Paris, France</i>	2022 – 2027

EXPERIENCE

Research internship — Laser beam propagation through the atmosphere <i>NCSR “Demokritos” · Athens, Greece</i> • Python simulation of laser propagation at 7 wavelengths (UV to CO₂), from 1 to 100 km • Modelling of scattering (Rayleigh, Mie), absorption (HITRAN) and turbulence • Wrote a detailed technical report (model, methods, results)	Jun. – Aug. 2026
Internship — 3D modelling of an orbital propellant depot <i>European Space Agency (ESA) · Paris, France</i> • Detailed Blender modelling of the components of an in-orbit refuelling concept, within a technical team	2024 · 6 weeks
Student job — Quality control in an industrial environment <i>Schneider Electric · Grenoble, France</i> • Quality control of finished products and verification of compliance with technical standards	2023 · 1 month

PROJECTS

Eco-design of the landing system of a reusable launcher — IPSA / CNES • Technical and economic feasibility study in a team of 6, supervised by CNES • Trade-off between 11 concepts, preliminary sizing, LCA, mass and cost budgets	2025 – 2026
Earth-Neptune mission • Jupiter gravity assist: patched-conic calculation, then simulation	STK / Astrogator
Stabilisation of an inverted pendulum (Segway) • Nonlinear model, cascaded PID with anti-windup, pole placement	MATLAB / Simulink
Satellite image classification with deep learning • Trained a ResNet network on the EuroSAT dataset to recognise land-cover types	Python

TECHNICAL SKILLS

Simulation & analysis	MATLAB / Simulink, STK / Astrogator, Python
Control	Discrete PID, state-space representation, pole placement, Kalman filter
Programming	Python, C / C++, VHDL / FPGA, Arduino
CAD & structures	CATIA V5, Creo, Blender, strength of materials

LANGUAGES & ACTIVITIES

Languages	French (native) · English B2 (TOEIC 830) · Spanish B2
Student club	IPSA F1 (2023 – 2026): treasurer, head of design, chassis assembly